

Rami Abou-Shamalah

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Education

MS, Applied Data Science <i>University of Michigan</i>	2023-2025
BSc, Applied Mathematics <i>University of Western Ontario</i>	2016-2020

Experience

Data Scientist <i>Sander Geophysics</i>	June 2023 - Present <i>Ottawa, Canada</i>
- Built end-to-end parallelized data pipeline that extracts historical projects from on-premise servers, fetching data within custom databases, performs QA/QC, removes outliers, normalizes the data, and engineers features - automating decades worth of training data for machine learning training purposes. - Implemented time series anomaly detection unsupervised model (AnomalyTransformer) via Arxiv in PyTorch, incorporated WandB to track experiments in real-time and Hydra for configuration file management. Afterwards fed the results of this model into a supervised XGBoost model to classify anomalies. This model reduced processing time by over 85% with over 95% accuracy. - Developed Python programs leveraging NumPy and Pandas for conducting statistical analysis of flight data by assessing the quality against client specifications, resulting in \$10,000 in annual savings.	
Data Analyst / Engineer <i>IStop</i>	June 2022 - June 2023 <i>Remote in Canada</i>
- Provided executives with a centralized dashboard for real-time, data-driven decision making, architecting and deploying a data warehouse in Snowflake, consolidating disparate sources including: payroll, ad spend, web analytics, and revenue streams. - Optimized caching and automated ETL flows from Firebase into Supase into IndexedDB (via REST APIs), allowing caching of 500MB+ of data client-side, reducing API calls by 85%. - Built OKR dashboards to quantify marketing ROI volume for stakeholder meetings, directly proving ad spend increases appointments volume. - Developed a revenue forecasting model in Python using historical data, achieving 85% accuracy. - Embedded Google Maps API to enable geospatial analysis, guiding city-level marketing decisions. - Maintained internal API documentation and built reusable RESTful endpoints for data ingestion.	
Data Analyst <i>Innovative Mining Solutions</i>	December 2021 – June 2022 <i>London, Canada</i>
- Led team of 5 in data collection and curation process: overseeing collection, digitizing, cleaning, transforming, and implementing QA/QC, thus preparing high quality and ML-ready training data. - Engineered geospatial features using QGIS software boosting model precision by ~22%. - Hosted training datasets on cloud-based PostgreSQL infrastructure with PostGIS optimization - Utilized PyTorch to train CNN on geophysical images, improving model accuracy by 40% - Presented domain-specifics for non-technical executives in company-wide meetings.	

Projects

Geospatial Prospectivity AI Platform	2025
- Designed a geospatial AI platform for mineral prospectivity using CNN-based analysis on geophysical raster datasets (gravity, magnetic). Trained deep models to identify mineral potential hotspots. - Served user-selected regions via Supabase API, using spatial queries to dynamically return results. Try the Interactive Model	